Mathematical Foundations of Computational Electrophysiology

A theorem-driven companion to CON^{2}PHYS

CON^{2}PHYS study notes

Revised working edition — October 2026

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How to read this book

We begin with ordinary sentences. The claim *Every spike belongs to a recorded neuron* can be true, false, or ambiguous if the meaning of *belongs* is not specified. Mathematics begins by removing ambiguity: we give names to objects, describe relations among them, and state exactly when a claim follows from its assumptions.

A *set* is a collection of objects. A *relation* describes which ordered pairs satisfy a condition. A *function* assigns one output to each admissible input. A signal is a function of time; a spike train can be represented by a counting measure; a covariance matrix represents pairwise relations among random variables; a neural trajectory is a function whose values lie in a vector space.

How to use the numbered results. Results 1–92 form the original indexed catalogue of named theorems, lemmas, inequalities, criteria and identities. The revised curriculum interleaves rigorous chapters on measures, Gaussian conditioning, electrode biophysics, digital filters, GLMs, renewal spectra, population identifiability, causal inference and neural dynamics. Classical sampling and random-matrix theorems are placed with their spectral prerequisites. The word *result* includes some identities and principles rather than only formal theorems. Compact catalogue entries do not replace proofs.

Notation. We write $x \in A$ for membership, $A \subseteq B$ for inclusion, $f : A \to B$ for a function, $\mathbb{E}[X]$ for expectation, $\mathrm{Cov}(X, Y)$ for covariance, and X^* for conjugate transpose.

An example before advanced mathematics. Let $U = \{u_1, u_2, u_3\}$ be the set of identified units and let $T = [0, 10)$ seconds. A spike event is an ordered pair $(u, t) \in U \times T$. The relation $R \subseteq U \times T$ records which spikes occurred. A count function maps each unit $u \in U$ to its number of spike times in T. This is the progression from a sentence to sets, relations, functions, numbers, and finally a statistical model.

Scientific caution. The repository's `arsenal/README.md` explicitly describes its MATLAB methods as exploratory and not validated for scientific inference. The mathematical validity of a formula is separate from the adequacy of the assumptions required to infer physiology from data.

Reading order after the review. The physical forward problem—membrane current to extracellular potential to digitized signal—now precedes its spectral and inferential models. For a chapter-by-chapter accounting of resolved and outstanding gaps, see the revision matrix linked in the LaTeX README. A complete proof of every theorem is not claimed.

Part I

Language and mathematical foundations

Chapter 1

From ordinary statements to mathematical structures

A mathematical proof is a sequence of statements whose conclusion follows from explicitly declared assumptions. Start with the sentence *If a spike occurred in an interval, the interval contains a spike.* The implication is meaningful only after the interval and the membership relation have been defined.

Definition 1.1. A *relation* from A to B is a subset $R \subseteq A \times B$. It is a *function* $f : A \to B$ when every $a \in A$ occurs as the first coordinate of exactly one pair $(a, b) \in R$.

For example, a spike train of a unit is a subset of its observation time interval. A classification of spike waveforms by cluster labels is a function only when every input waveform receives exactly one label. This does not imply that the label identifies a physiological neuron.

Result 1: De Morgan's laws. *Precise setting:* For propositions P, Q, $\neg(P \wedge Q) \Leftrightarrow (\neg P) \vee (\neg Q)$; for sets, $(A \cup B)^c = A^c \cap B^c$, relative to an ambient set.

Why it matters here: Transform selection criteria and complements of detected event sets.

Result 2: Quantifier duality. *Precise setting:* For any predicate P, $\neg(\forall x\, P(x)) \Leftrightarrow \exists x\, \neg P(x)$; likewise $\neg(\exists x\, P(x)) \Leftrightarrow \forall x\, \neg P(x)$.

Why it matters here: Express universal reliability claims and their counterexamples.

Result 3: Mathematical induction (principle). *Precise setting:* If $P(0)$ holds and $P(n) \Rightarrow P(n+1)$ for $n \in \mathbb{N}$, then $P(n)$ holds for all $n \in \mathbb{N}$.

Why it matters here: Prove recurrences, finite sums, binning bounds, and iterative invariants.

Result 4: Recursion theorem on the natural numbers. *Precise setting:* With a starting value and a step map, there is a unique sequence satisfying the corresponding recursive definition in the standard set-theoretic setting.

Why it matters here: Construct discrete filters, numerical solvers, and event counters.

Result 5: Equivalence relations and partitions. *Precise setting:* Reflexive, symmetric, transitive relations correspond to partitions into disjoint nonempty equivalence classes.

Why it matters here: Distinguish clustering as a partition from physiological identification.

Result 6: Well-ordering principle. *Precise setting:* Every nonempty subset of $\mathbb{N}$ has a least element.

Why it matters here: Prove termination of monotone integer-index algorithms.

Result 7: Cantor–Schroeder–Bernstein theorem. *Precise setting:* If injections $A \to B$ and $B \to A$ exist, then a bijection $A \to B$ exists.

Why it matters here: Understand cardinality and information-preserving relabelings.

Result 8: Cantor's diagonal argument. *Precise setting:* There is no surjection $\mathbb{N} \to (0, 1)$; the real interval is uncountable.

Why it matters here: Understand the distinction between finite sampled arrays and continuum models.

Result 9: Least-upper-bound property (completeness axiom for $\mathbb{R}$). *Precise setting:* Every nonempty subset of $\mathbb{R}$ bounded above has a real supremum.

Why it matters here: Build limits, continuity, completeness, and convergence of numerical approximations.

Result 10: Archimedean property and density of $\mathbb{Q}$. *Precise setting:* For real x there is an integer $n > x$; between any $a < b$ in $\mathbb{R}$ lies some $q \in \mathbb{Q}$.

Why it matters here: Approximate continuous times and amplitudes by rational values, without confusing approximation with exact sampling.

Example 1.2 (Equivalence classes of spike timestamps)**.** For a fixed bin width $\Delta > 0$, let $s \sim t$ when $\lfloor s/\Delta \rfloor = \lfloor t/\Delta \rfloor$. This relation is reflexive, symmetric, and transitive. Each equivalence class represents one temporal bin, not one spike. A spike-count sequence counts members of each class.

Interpretation warning 1.3. Membership ($x \in A$), inclusion ($A \subseteq B$), and association under a binary relation (xRy) are distinct concepts. An observed cross-correlation is not evidence that a causal or anatomical edge exists.

Chapter 2

The measure-theoretic vocabulary behind probability

A probability model is not merely a formula for averages. It states *which outcomes exist*, *which events can be assigned probabilities*, and *which calculations are allowed*. This chapter provides the missing hypotheses for Results 21–24, 40–46, and 71–76.

Definition 2.1 (Sigma-algebra and measure)**.** For a set Ω, a σ-algebra $\mathcal{F}$ is a family of subsets containing Ω, closed under complements and countable unions. A measure $\mu : \mathcal{F} \to [0, \infty]$ satisfies $\mu(\varnothing) = 0$ and countable additivity over disjoint events. A probability measure has $\mu(\Omega) = 1$. A measure is *σ-finite* if Ω is a countable union of measurable sets of finite measure.

Definition 2.2 (Random variable and expectation)**.** A real-valued random variable X is a measurable map $(\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$: for every Borel set B, its preimage $X^{-1}(B)$ is an event. If $\int |X| \, d\mathbb{P} < \infty$, define $\mathbb{E}X = \int X \, d\mathbb{P}$. The space L^2 consists of equivalence classes of variables with $\mathbb{E}|X|^2 < \infty$; equality almost surely is identified. Its inner product is $\langle X, Y \rangle = \mathbb{E}[X\overline{Y}]$. A *Hilbert space* is a complete inner-product space; L^2 is one. Completeness is what makes orthogonal projection possible.

Proposition 2.3 (Conditional expectation is a projection)**.** *For $X \in L^2$ and a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$, the conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is characterized by being $\mathcal{G}$-measurable and satisfying $\mathbb{E}[(X - \mathbb{E}[X \mid \mathcal{G}])Z] = 0$ for all bounded $\mathcal{G}$-measurable Z. It is the orthogonal projection of X onto the closed subspace $L^2(\mathcal{G})$.*

Proof. The defining conditional-expectation identity gives orthogonality for bounded Z. Truncation followed by L^2 approximation extends it to $Z \in L^2(\mathcal{G})$. The projection theorem then characterizes the minimizer of $\mathbb{E}[(X - Z)^2]$ among such Z. □

This explains why a conditional firing-rate prediction is an optimal mean-square prediction *relative to the variables conditioned upon*, not relative to every unobserved brain state.

Definition 2.4 (Stationarity, ergodicity, and predictability)**.** A process X_t is *wide-sense stationary* if its mean is constant and $\mathrm{Cov}(X_t, X_s)$ depends only on $t - s$. It is *strictly stationary* if all finite-dimensional joint laws are time-shift invariant. An invariant process is *ergodic* when invariant events have probability zero or one. A predictable intensity $\lambda(t \mid \mathcal{H}_{t-})$ is measurable relative to information available *just before* t; this forbids predicting a spike using its own occurrence. The time-integrated intensity is called its *compensator* when the Doob–Meyer conditions are satisfied. For a regular parametric model p_θ, the *score* is

$U_\theta = \nabla_\theta \log p_\theta(X)$ and *Fisher information* is $\mathcal{I}(\theta) = \mathbb{E}_\theta[U_\theta U_\theta^\top]$, under assumptions justifying its finiteness and calculus.

Proposition 2.5 (Law of total variance revisited)**.** *For $X \in L^2$,*

$$\operatorname{Var}(X) = \mathbb{E}[\operatorname{Var}(X \mid \mathcal{G})] + \operatorname{Var}(\mathbb{E}[X \mid \mathcal{G}]).$$

Proof. Write $X - \mathbb{E}X = (X - M) + (M - \mathbb{E}X)$ where $M = \mathbb{E}[X \mid \mathcal{G}]$. The cross term vanishes by orthogonality. Taking squared expectations yields the identity. □

Example 2.6 (LFP recording versus ensemble)**.** If a stationary and ergodic model describes a long LFP trace, its time average can estimate the ensemble mean under integrability. Stationarity alone does not justify equating those means, nor does increasing the sample rate create independent trials.

Chapter 3

Real analysis, inequalities, and integration

Before manipulating signals, one must specify their domains, continuity, integrability, and normed spaces. A sampled waveform is a finite vector; a continuous-time waveform belongs to a function class. The two are connected through sampling hypotheses, not merely by notation.

Result 11: Triangle inequality. *Precise setting:* In a normed space $\|x + y\| \leq \|x\| + \|y\|$.

Why it matters here: Bound approximation and reconstruction errors.

Result 12: Cauchy–Schwarz inequality. *Precise setting:* In an inner-product space $|\langle x, y\rangle| \leq \|x\|\|y\|$; for square-integrable random variables, $|\operatorname{Cov}(X, Y)|^2 \leq \operatorname{Var}(X) \operatorname{Var}(Y)$.

Why it matters here: Bound Pearson correlation, coherence, and inner-product decoders.

Result 13: Holder's inequality. *Precise setting:* For $1/p + 1/q = 1$ with $p, q \geq 1$, and $f \in L^p$, $g \in L^q$, $\int |fg| \leq \|f\|_p \|g\|_q$.

Why it matters here: Establish existence of products and moments.

Result 14: Minkowski's inequality. *Precise setting:* For $p \geq 1$, $\|f + g\|_p \leq \|f\|_p + \|g\|_p$.

Why it matters here: Justify L^p norms on signals and residuals.

Result 15: Bolzano–Weierstrass theorem. *Precise setting:* Every bounded sequence in finite-dimensional Euclidean space has a convergent subsequence.

Why it matters here: Understand compactness arguments in parameter estimation.

Result 16: Heine–Borel theorem. *Precise setting:* A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.

Why it matters here: Provide conditions for existence of finite-dimensional constrained optima.

Result 17: Extreme value theorem. *Precise setting:* Continuous functions on nonempty compact sets attain their minimum and maximum.

Why it matters here: Show that certain constrained model-fitting objectives have minimizers.

Result 18: Mean value theorem. *Precise setting:* If f is continuous on $[a, b]$ and differentiable on (a, b), then some $c \in (a, b)$ satisfies $f(b) - f(a) = f'(c)(b - a)$.

Why it matters here: Control local approximation and numerical differentiation errors.

Result 19: Taylor's theorem with remainder. *Precise setting:* Under requisite differentiability, a function equals its order-k Taylor polynomial plus a controlled remainder.

Why it matters here: Linearize nonlinear neural state equations near operating points.

Result 20: Fundamental theorem of calculus. *Precise setting:* For continuous f, $F(x) = \int_a^x f(t)\,dt$ satisfies $F'(x) = f(x)$; suitable primitives recover integrals.

Why it matters here: Relate firing intensities to integrated expected event counts.

Result 21: Dominated convergence theorem. *Precise setting:* If measurable $f_n \to f$ almost everywhere and $|f_n| \le g$ for integrable g, then $\int f_n \to \int f$.

Why it matters here: Justify exchanging controlled limits and expectations.

Result 22: Monotone convergence and Fatou's lemma. *Precise setting:* For nonnegative measurable increasing $f_n \uparrow f$, $\int f_n \uparrow \int f$; Fatou gives $\int \liminf f_n \le \liminf \int f_n$.

Why it matters here: Handle nonnegative measures and event-count approximations.

Result 23: Radon–Nikodym theorem. *Precise setting:* If ν is a sigma-finite measure absolutely continuous relative to a sigma-finite measure μ, there is a measurable density $d\nu/d\mu$.

Why it matters here: Define likelihood and probability density correctly.

Result 24: Banach fixed-point theorem. *Precise setting:* A contraction on a nonempty complete metric space has exactly one fixed point, reached by iteration from any starting point.

Why it matters here: Justify convergence of certain recursive estimators under contraction.

Definition 3.1. A sequence x_n converges to x in a metric space if for every $\epsilon > 0$ there is an N such that $n \ge N$ implies $d(x_n, x) < \epsilon$. Convergence in probability, almost surely, and in L^2 have different definitions; none should silently replace another.

Interpretation warning 3.2. The derivative of an estimated signal can amplify measurement noise. A valid derivative theorem for a smooth function does not prove that a finite-difference estimate of noisy LFP samples is accurate.

Part II

Vector spaces and statistical inference

Chapter 4

Vector spaces, matrices, and high-dimensional geometry

Let $X \in \mathbb{R}^{T \times N}$ collect N neural variables across T observations. Its columns may be neurons, channels, or features; centering across rows changes the statistical quantity being modeled. Linear algebra does not specify how trials should be aligned or how observations should be validated.

Result 25: Rank–nullity theorem. *Precise setting:* For linear $A : V \to W$ with $\dim V < \infty$, $\dim \ker A + \operatorname{rank} A = \dim V$.

Why it matters here: Count identifiable and unidentifiable latent directions.

Result 26: Orthogonal projection theorem. *Precise setting:* In a Hilbert space every element has a unique closest point in a closed subspace; the residual is orthogonal to that subspace.

Why it matters here: Derive least squares and orthogonal signal decompositions.

Result 27: Spectral theorem. *Precise setting:* A real symmetric matrix admits an orthonormal eigenbasis and real eigenvalues; a covariance matrix has nonnegative eigenvalues.

Why it matters here: Explain the eigenvectors underlying population PCA.

Result 28: Singular value decomposition. *Precise setting:* Every real matrix X admits $X = U\Sigma V^\top$ with orthogonal factors and nonnegative singular values.

Why it matters here: Implement low-rank population representations.

Result 29: Eckart–Young–Mirsky theorem. *Precise setting:* Truncating SVD to rank k gives an optimal rank-k approximation in Frobenius and spectral norms, more generally unitarily invariant norms.

Why it matters here: Give a reconstruction criterion for reduced-dimensional activity.

Result 30: Courant–Fischer min–max theorem. *Precise setting:* Eigenvalues of a symmetric matrix have variational characterizations over subspaces via Rayleigh quotients.

Why it matters here: Derive why sequential PCA directions maximize remaining variance.

Result 31: Gauss–Markov theorem. *Precise setting:* With a full-column-rank linear model, zero-mean errors with common variance and no correlation, OLS is BLUE.

Why it matters here: Clarify which optimality claim linear decoders can and cannot make.

Result 32: Schur complement identities. *Precise setting:* For an invertible block, Schur complements characterize positive definiteness and block inverses under the stated conditions.

Why it matters here: Compute conditional Gaussian covariance and latent uncertainty.

Result 33: Weyl eigenvalue inequalities. *Precise setting:* For Hermitian A, E, sorted eigenvalues of $A + E$ differ from those of A by at most $\|E\|_2$.

Why it matters here: Control eigenspectrum changes caused by covariance perturbations.

Result 34: Davis–Kahan sin-theta theorem. *Precise setting:* For symmetric matrices with separated spectral clusters, their invariant subspaces have angle bounds proportional to perturbation size divided by a suitable eigengap.

Why it matters here: Assess when estimated neural subspaces are stable.

Result 35: Sherman–Morrison–Woodbury identity. *Precise setting:* For invertible A, C and required inverse, $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$.

Why it matters here: Speed covariance updates when their rank is small.

Result 36: Matrix determinant lemma. *Precise setting:* For invertible A, C, $\det(A + UCV) = \det(C^{-1} + VA^{-1}U)\det(C)\det(A)$.

Why it matters here: Accelerate Gaussian latent-variable likelihood evaluations.

Result 37: Mercer's theorem. *Precise setting:* For a continuous positive-semidefinite kernel on a compact domain with a suitable finite reference measure, an orthogonal eigenfunction expansion exists under its hypotheses.

Why it matters here: Understand kernel feature representations in GP modeling.

Result 38: Karhunen–Loeve theorem. *Precise setting:* A second-order process admits a mean-square orthogonal expansion in covariance eigenfunctions under suitable regularity.

Why it matters here: Link temporal covariance to latent trajectories and function-space PCA.

Result 39: Kruskal's uniqueness criterion for CP tensors. *Precise setting:* A canonical polyadic tensor decomposition is essentially unique if its factor Kruskal ranks meet an appropriate sufficient bound, e.g. $k_A + k_B + k_C \geq 2R + 2$ for three factors.

Why it matters here: Reason about identifiability in trial-by-time-by-neuron decompositions.

Definition 4.1. For covariance $C \succeq 0$ with eigenvalues λ_i, the participation ratio is

$$D_{\mathrm{PR}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

It measures effective spectral dimensionality, not the intrinsic manifold dimension without further assumptions.

Interpretation warning 4.2. Truncated SVD solves a matrix approximation problem; recovering biological latent variables is an additional inference problem. Coherent noise, trial structure, and preprocessing can dominate the covariance eigenspectrum.

Chapter 5

Probability, estimation, and valid inference

A random variable is a measurable function on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. An expectation is an integral with respect to $\mathbb{P}$. The distinction between a deterministic signal and a stochastic process must be preserved even when both are stored as the same array of samples.

Result 40: Bayes' theorem and total probability. *Precise setting:* For events with positive conditioning probability, $\mathbb{P}(A \mid B) = \mathbb{P}(B \mid A)\mathbb{P}(A)/\mathbb{P}(B)$; for a partition (A_i), $\mathbb{P}(B) = \sum_i \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)$.

Why it matters here: Build probabilistic decoders and posterior state estimates.

Result 41: Law of total expectation. *Precise setting:* For integrable Y, $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid G]]$.

Why it matters here: Marginalize trial-to-trial latent gain or unobserved state.

Result 42: Law of total variance. *Precise setting:* For square-integrable Y, $\mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y \mid G)] + \mathrm{Var}(\mathbb{E}[Y \mid G])$.

Why it matters here: Explain Poisson noise plus variable excitability.

Result 43: Law of total covariance. *Precise setting:* For square-integrable X, Y, $\mathrm{Cov}(X, Y) = \mathbb{E}[\mathrm{Cov}(X, Y \mid G)] + \mathrm{Cov}(\mathbb{E}[X \mid G], \mathbb{E}[Y \mid G])$.

Why it matters here: Separate residual noise correlation from shared modulating factors.

Result 44: Law of large numbers. *Precise setting:* Sample means converge to the expectation under appropriate independence or dependence, moment, and ergodicity assumptions.

Why it matters here: Explain consistency and why many time bins need not imply many independent samples.

Result 45: Central limit theorem. *Precise setting:* For iid observations with finite nonzero variance, normalized sample-mean fluctuations converge in distribution to a normal law; dependent-data versions need stronger conditions.

Why it matters here: Motivate uncertainty quantification for population statistics.

Result 46: Birkhoff ergodic theorem. *Precise setting:* For an integrable observable of a

measure-preserving system, time averages converge almost surely to a conditional invariant expectation; ergodicity makes the limit equal the ensemble expectation.

Why it matters here: Clarify when one long recording can replace independent replicates.

Result 47: Markov and Chebyshev inequalities. *Precise setting:* For $X \geq 0$, $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$; for finite variance, $\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \mathrm{Var}(X)/a^2$.

Why it matters here: Derive conservative uncertainty bounds for estimators.

Result 48: Hoeffding and Bernstein concentration bounds. *Precise setting:* Bound sums of independent bounded variables, or those obeying specified moment constraints; extensions to dependence require care.

Why it matters here: Control finite-sample false alarms under valid independence assumptions.

Result 49: Delta method. *Precise setting:* A differentiable map of an asymptotically normal estimator is asymptotically normal after suitable scaling.

Why it matters here: Approximate uncertainty of nonlinear indices.

Result 50: Cramer–Rao inequality. *Precise setting:* With regular parametric likelihood and unbiased estimator, variance is bounded by reciprocal Fisher information (or its multivariate matrix counterpart).

Why it matters here: Compare theoretical accuracy limits of firing-rate estimators.

Result 51: Neyman–Pearson lemma. *Precise setting:* For testing one simple hypothesis against another, thresholding the likelihood ratio yields a most-powerful test at fixed size, allowing randomization at a threshold.

Why it matters here: Understand what an optimal detection test does and does not guarantee.

Result 52: Wilks' theorem. *Precise setting:* Under regularity and appropriate interior-point assumptions, twice a nested-model log-likelihood difference is asymptotically chi-squared with degrees of freedom equal to parameter difference.

Why it matters here: Test nested GLM or autoregressive restrictions when assumptions apply.

Result 53: Fisher–Neyman factorization theorem. *Precise setting:* In dominated statistical families, a statistic is sufficient precisely when the likelihood has an appropriate factorization.

Why it matters here: Justify reductions of data into counts only for particular models.

Result 54: Exact randomization testing under exchangeability. *Precise setting:* If the null distribution is invariant under a finite transformation group, a correctly ranked test statistic gives a valid finite-sample randomization test.

Why it matters here: Build scientifically matched permutation and jitter controls.

Result 55: Benjamini–Hochberg FDR control. *Precise setting:* For valid p-values under independence or certain positive dependence assumptions, the step-up procedure controls expected false discovery proportion.

Why it matters here: Correct across electrodes and spectral comparisons.

Result 56: Isserlis' or Wick's Gaussian moment theorem. *Precise setting:* Centered jointly Gaussian even-order moments decompose into sums over pairings of covariances; odd-order moments vanish.

Why it matters here: Evaluate Gaussian baseline moments and covariance estimators.

Example 5.1 (A modulated Poisson count)**.** Let $N \mid G \sim \text{Poisson}(\mu G)$, $G > 0$, and $\mathbb{E}G = 1$. The law of total variance gives

$$\mathbb{E}N = \mu, \qquad \text{Var}(N) = \mu + \mu^2 \,\text{Var}(G), \qquad \frac{\text{Var}(N)}{\mathbb{E}N} = 1 + \mu \,\text{Var}(G).$$

This derivation explains overdispersion without claiming that a particular dataset actually follows this latent-gain model. See Goris et al. [13].

Interpretation warning 5.2. A confidence interval does not automatically have its nominal coverage after choosing frequency bands, time windows, and hypotheses from the same data without accounting for selection.

Chapter 6

A working toolkit for Gaussian models

Gaussian conditioning silently appears in Kalman filters, Gaussian processes, factor analysis, and several multivariate estimators. It belongs before relying on their formulas.

Definition 6.1 (Multivariate normal vector)**.** A vector $X \in \mathbb{R}^d$ is jointly Gaussian if every scalar linear combination $a^\top X$ is normally distributed (possibly degenerate). Its mean μ and covariance $\Sigma \succeq 0$ characterize its distribution. When $\Sigma \succ 0$, its density is

$$p(x) = (2\pi)^{-d/2}|\Sigma|^{-1/2}\exp\left[-\frac{1}{2}(x-\mu)^\top\Sigma^{-1}(x-\mu)\right].$$

Theorem 6.2 (Gaussian conditioning by a Schur complement)**.** *Suppose (X, Y) is jointly normal and partition its mean and covariance into μ_X, μ_Y and blocks Σ_{XX}, Σ_{XY}, Σ_{YY}. If Σ_{YY} is invertible,*

$$\mathbb{E}[X \mid Y = y] = \mu_X + \Sigma_{XY}\Sigma_{YY}^{-1}(y - \mu_Y),$$
$$\mathrm{Cov}(X \mid Y) = \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}.$$

Proof. Define $R = X - \mu_X - \Sigma_{XY}\Sigma_{YY}^{-1}(Y - \mu_Y)$. Then $\mathrm{Cov}(R, Y) = 0$. Joint Gaussianity makes R and Y independent. Expanding $\mathrm{Cov}(R)$ gives the Schur complement. □

Proposition 6.3 (Wishart sample-covariance model)**.** *If $Z_1, \dots, Z_T$ are independent $\mathcal{N}_d(0, \Sigma)$ with known zero mean, then $\sum_{t=1}^{T} Z_t Z_t^\top$ has a Wishart distribution with scale Σ and T degrees of freedom. The usual centered sample covariance instead involves $T - 1$ degrees of freedom for independent Gaussian samples.*

This distributional assumption is the model behind classical random-covariance results; the same formula does not describe arbitrarily correlated neuronal time bins.

Proposition 6.4 (Multivariate Stein identity)**.** *For $X \sim \mathcal{N}_d(\mu, \Sigma)$ and a differentiable function f with appropriate integrability and boundary decay,*

$$\mathbb{E}[(X - \mu)f(X)] = \Sigma\,\mathbb{E}[\nabla f(X)].$$

Proof. For a nondegenerate Gaussian density p, $\nabla p(x) = -\Sigma^{-1}(x - \mu)p(x)$. Integrate $\nabla(fp)$ and use the vanishing boundary term. □

6.1 Gaussian KL divergence

For full-rank Gaussian distributions $P = \mathcal{N}(\mu_0, \Sigma_0)$ and $Q = \mathcal{N}(\mu_1, \Sigma_1)$ in dimension d,

$$D_{\mathrm{KL}}(P\|Q) = \frac{1}{2}\left[\operatorname{tr}(\Sigma_1^{-1}\Sigma_0) + (\mu_1 - \mu_0)^\top \Sigma_1^{-1}(\mu_1 - \mu_0) - d + \log\frac{\det\Sigma_1}{\det\Sigma_0}\right].$$

A proof expands $\mathbb{E}_P[\log p(X) - \log q(X)]$, then uses the covariance trace identity. The formula requires positive-definite covariance matrices; singular distributions need measure-theoretic treatment of their supports.

Example 6.5 (Neural latent variables)**.** A Kalman measurement update is repeated conditioning of a joint Gaussian state–observation vector. Its familiar gain matrix is therefore a Schur-complement calculation, not a property of all nonlinear or non-Gaussian observation processes.

Part III

Physical and digital origins of neural measurements

Chapter 7

From membrane currents to extracellular electrodes

An extracellular electrode does not measure a neuron's voltage in isolation. It measures a difference of electric potential created by spatially distributed ionic currents, tissue geometry, conductivity, referencing, and acquisition electronics [5, 11].

7.1 Electrochemical driving forces

For an ideal dilute permeant ion with valence $z \neq 0$, equality of electrochemical potentials on the two sides of a membrane gives

$$E_{\text{ion}} = \frac{RT}{zF} \log \frac{c_{\text{out}}}{c_{\text{in}}}.$$

This is the Nernst equilibrium potential under its concentration and temperature assumptions. The Goldman–Hodgkin–Katz formula extends the constant-field model to multiple monovalent ionic species with permeability parameters; it is not equivalent to measuring those permeabilities from a single LFP. In the Hodgkin–Huxley description the sodium current is modeled as $I_{\text{Na}} = \bar{g}_{\text{Na}} m^3 h (V - E_{\text{Na}})$; gating variables obey voltage-dependent kinetic equations. The passive cable approximation has the schematic form

$$c_m \partial_t V(x,t) = g_a \partial_{xx} V(x,t) - g_m (V - E_L) + i_{\text{ext}}(x,t),$$

with coefficients per unit length and consistent units. Cable and membrane models determine transmembrane sources; they are not by themselves electrode forward models.

7.2 Quasistatic field equation

At electrophysiological frequencies and spatial scales where inductive effects are negligible, write $\mathbf{E} = -\nabla\phi$. Let impressed current $\mathbf{J}_p$ and Ohmic return current $-\boldsymbol{\sigma}\nabla\phi$ sum to a divergence-free total current. Current conservation implies

$$\nabla\cdot(\boldsymbol{\sigma}\nabla\phi) = \nabla\cdot\mathbf{J}_p, \quad \text{or} \quad -\nabla\cdot(\boldsymbol{\sigma}\nabla\phi) = q, \quad q = -\nabla\cdot\mathbf{J}_p.$$

The tensor $\boldsymbol{\sigma}$ may be anisotropic and spatially inhomogeneous. To solve the equation, the domain, conductivity, boundary conditions and electrode reference must be specified. For an

infinite isotropic medium, a point source of strength I located at $\mathbf{r}_0$ gives, away from that point,

$$\phi(\mathbf{r}) = \frac{I}{4\pi\sigma|\mathbf{r} - \mathbf{r}_0|}.$$

Superposition yields a discrete source model $\mathbf{v} = G\mathbf{s}$ when the forward physics is linear, with lead-field entries obtained by integrating the Green kernel over chosen source basis functions. A dipole arises as the leading far-field contribution when net source current cancels; line-source models integrate along a dendrite.

Proposition 7.1 (Uniqueness for fixed Dirichlet conditions)**.** *Let Ω be a bounded connected region and $\boldsymbol{\sigma}$ uniformly positive definite. If two weak solutions of the same elliptic equation have equal Dirichlet boundary values, they agree (almost everywhere).*

Proof. Their difference w satisfies the homogeneous equation and zero boundary values. Multiply by w, integrate by parts, and use the boundary condition: $\int_\Omega \nabla w \cdot \boldsymbol{\sigma} \nabla w \, dx = 0$. Positivity implies $\nabla w = 0$; the boundary forces $w = 0$. For pure Neumann conditions, uniqueness holds only up to an additive constant, with a compatibility condition on source flux. □

7.3 Why the inverse need not be unique

Suppose an electrode array measures $\mathbf{v} = G\mathbf{s}$. If $G\mathbf{h} = 0$ for nonzero $\mathbf{h}$, then $G(\mathbf{s} + \mathbf{h}) = G\mathbf{s}$. This is exact finite-dimensional nonidentifiability. In the continuum, smoothing operators can also be ill-conditioned or compact, so tiny measurement errors may cause enormous source errors even when uniqueness holds in a restricted model. *Hadamard well-posedness* requires existence, uniqueness and continuous dependence of solutions on data. A linear operator is *compact* if it maps bounded sets into sets with compact closure; its singular values in infinite-dimensional inverse problems can converge to zero. For singular system $Gv_j = \sigma_j u_j$ and data b, the formal inverse contains $\langle b, u_j \rangle / \sigma_j$ and magnifies noise when $\sigma_j \to 0$. A classical Picard-type condition controls the squared normalized coefficients for compatible data. Tikhonov regularization instead solves

$$\widehat{\mathbf{s}}_\lambda = \arg\min_{\mathbf{s}} \{\|G\mathbf{s} - \mathbf{v}\|_2^2 + \lambda \|\mathbf{s}\|_2^2\} = (G^\top G + \lambda I)^{-1} G^\top \mathbf{v}.$$

Example 7.2 (MATLAB and interpretation)**.** `field_potential_forward.m` receives G; it does *not* compute tissue conductivity or the Green kernel. `field_potential_inverse_tikhonov.m` returns one regularized source estimate, not a unique membrane-current pattern. `current_source_density.m` needs spatial geometry and differentiated potentials; references and volume conduction remain crucial.

Chapter 8

Digital filtering, stability, and aliasing

The first operation in many LFP workflows is band-pass filtering. Its interpretation requires a distinction between a sampled sequence, its discrete-time filter, and the underlying analog field [16].

8.1 Poisson summation and aliasing

For suitable rapidly decreasing $x(t)$, Fourier convention $X(f) = \int x(t)e^{-2\pi ift}\,dt$, and sampling frequency F_s, Poisson summation gives the periodic spectrum of its samples:

$$\sum_{n\in\mathbb{Z}} x(n/F_s)e^{-2\pi inf/F_s} = F_s \sum_{k\in\mathbb{Z}} X(f + kF_s).$$

The observed spectrum is thus a *sum of translated copies*. The copies overlap unless the analog spectrum is sufficiently bandlimited; an anti-alias filter must act *before* digitization. As a concrete counterexample, the sequences from frequencies f and $f + F_s$ agree because $e^{2\pi i(f+F_s)n/F_s} = e^{2\pi ifn/F_s}$. No post-acquisition algorithm can distinguish them without additional prior information.

8.2 Convolution, the z-transform, and poles

For a causal linear time-invariant discrete filter, $y[n] = \sum_{k=0}^{\infty} h[k]x[n-k]$, with

$$H(z) = \sum_{k=0}^{\infty} h[k]z^{-k}.$$

Its frequency response is $H(e^{i\omega})$ when that boundary value exists. Absolute summability $\sum_k |h[k]| < \infty$ is sufficient for bounded-input bounded-output (BIBO) stability. For rational causal filters in minimal form, BIBO stability requires all poles strictly inside the unit circle.

Proposition 8.1 (First-order recursive filter)**.** *For $y[n] = ay[n-1] + x[n]$ with zero initial condition, $h[k] = a^k\mathbf{1}_{k\geq 0}$. The system is BIBO stable if and only if $|a| < 1$.*

Proof. The absolute impulse-response sum is $\sum_{k\geq 0} |a|^k = (1-|a|)^{-1}$ when $|a| < 1$. If $|a| \geq 1$, the impulse response is not absolutely summable; there exists a bounded input producing an unbounded output. □

8.3 FIR versus IIR; delay and zero-phase processing

An FIR filter has a finite impulse response and is BIBO stable. An IIR filter may have sharper transitions for a given order, but its poles and finite-precision implementation need inspection. For nonzero frequency response, define group delay $\tau_g(\omega) = -d \arg H(e^{i\omega})/d\omega$. A forward-backward filter ideally has response $H(e^{i\omega})H(e^{-i\omega}) = |H(e^{i\omega})|^2$ for real filter coefficients. This suppresses phase delay but is *noncausal*, changes the effective magnitude response, and requires boundary extrapolation: edge transients are not removed. An ideal brick-wall filter has an infinitely long sinc-shaped impulse response; abrupt truncation creates Gibbs-type ringing.

Example 8.2 (What the arsenal computes)**.** `bandpass_lfp.m` designs a Butterworth bandpass, converts it to second-order sections and applies `filtfilt`. The returned trace is a zero-phase *offline estimate*, not the output of a causal online instrument. The initial and final samples need explicit transient inspection. Filtering cannot remove volume conduction, spike-sorting errors or physiological common inputs.

Part IV

Spectra, spike trains, and neural dynamics

Chapter 9

Harmonic analysis and spectral time series

A continuous signal is a function $x : \mathbb{R} \to \mathbb{R}$ or $\mathbb{C}$. The Fourier transform, when meaningful in the appropriate sense, is $X(f) = \int_{\mathbb{R}} x(t)e^{-2\pi ift}\,dt$. A length-$N$ recording is a different object: a discrete sequence. An FFT of the recorded sequence does not directly reveal a unique continuous generating signal without sampling assumptions.

Result 57: Euler's formula and Fourier orthogonality. *Precise setting:* $e^{i\theta} = \cos\theta + i\sin\theta$ and $\int_0^T e^{2\pi i(k-\ell)t/T}\,dt = 0$ for distinct integers k, ℓ.

Why it matters here: Derive discrete spectral coefficients and phase representations.

Result 58: Parseval's identity and Plancherel's theorem. *Precise setting:* For appropriate Fourier expansions/transforms, energy in time equals energy in frequency, e.g. $\|x\|_2^2 = \|X\|_2^2$ under a unitary convention.

Why it matters here: Normalize power and check spectral energy.

Result 59: Fourier convolution theorem. *Precise setting:* For suitable integrable functions, $\mathcal{F}(f * g) = \mathcal{F}(f)\mathcal{F}(g)$, with conventions fixed.

Why it matters here: Relate a digital filter's impulse response to its transfer function.

Result 60: Fourier shift and modulation identities. *Precise setting:* Translation of a signal multiplies its spectrum by a complex phase; multiplication by a complex sinusoid translates frequency.

Why it matters here: Understand lags, phase offsets, and analytic demodulation.

Result 61: Heisenberg–Gabor uncertainty inequality. *Precise setting:* With standard centering and Fourier conventions, a nonzero finite-energy waveform has time spread and frequency spread whose product obeys a nonzero lower bound.

Why it matters here: Understand the temporal and spectral resolution trade-off.

Result 62: Paley–Wiener theorems. *Precise setting:* Appropriate compact-support and exponential-type conditions characterize special classes of Fourier transforms; compactly supported time-domain signals have entire Fourier transforms of controlled exponential growth under standard hypotheses. In particular, nonzero ordinary signals cannot have both time and frequency support compact: the identity theorem supplies this corollary.

Why it matters here: Explain the role of analyticity in exact localization; Benedicks-type results strengthen restrictions for finite-measure supports.

Result 63: Riemann–Lebesgue lemma. *Precise setting:* The Fourier transform of an L^1 function vanishes at infinity.

Why it matters here: Understand high-frequency decay under integrability assumptions.

Result 64: Bochner and Herglotz spectral representation. *Precise setting:* Continuous positive-definite functions on $\mathbb{R}$ are Fourier transforms of finite nonnegative measures; discrete-time analogues use nonnegative measures on the frequency circle.

Why it matters here: Characterize admissible stationary covariance functions and spectra.

Result 65: Cramer's spectral representation. *Precise setting:* A zero-mean weakly stationary second-order process can be represented via orthogonal-increment random spectral measures under the standard hypotheses.

Why it matters here: Interpret stochastic LFP spectra rather than deterministic FFTs alone.

Result 66: Wold decomposition. *Precise setting:* A purely nondeterministic covariance-stationary discrete-time process has a one-sided linear innovations representation; general stationary processes decompose into deterministic and purely nondeterministic parts.

Why it matters here: Understand assumptions behind linear innovations and VAR models.

Result 67: Yule–Walker equations. *Precise setting:* For a stationary autoregressive process, lagged covariance satisfies linear equations determined by AR coefficients.

Why it matters here: Relate autocovariance to AR spectral estimators.

Result 68: Wiener–Hopf normal equations. *Precise setting:* The least-squares optimal linear predictor has a residual orthogonal to the predictor space, yielding covariance normal equations.

Why it matters here: Derive linear decoding and linear prediction filters.

Result 69: Spectral coherence bound. *Precise setting:* A positive-semidefinite two-signal spectral matrix satisfies $|S_{xy}(f)|^2 \le S_{xx}(f)S_{yy}(f)$ where the spectra exist; hence $0 \le C_{xy}(f) \le 1$ for positive denominators.

Why it matters here: Interpret magnitude-squared LFP coherence.

Result 70: Hilbert-transform Fourier multiplier identity. *Precise setting:* On $L^2(\mathbb{R})$, with appropriate Fourier convention, $\mathcal{F}(\mathcal{H}x)(f) = -i\,\mathrm{sgn}(f)\mathcal{F}x(f)$.

Why it matters here: Build analytic signals and interpret instantaneous phase.

Definition 9.1. For zero-mean jointly wide-sense stationary signals x, y, let $R_{xy}(\tau) = \mathbb{E}[x(t+\tau)\overline{y(t)}]$. Under appropriate integrability, $S_{xy} = \mathcal{F}R_{xy}$. Magnitude-squared coherence is $C_{xy} = |S_{xy}|^2/(S_{xx}S_{yy})$ wherever the denominator is strictly positive.

Interpretation warning 9.2. Coherence is spectral dependence, not anatomical connectivity. Common reference, volume conduction, leakage, sample-size bias, and nonstationarity can distort interpretation [4, 8].

Chapter 10

Time–frequency representations and estimator uncertainty

A discrete Fourier transform can be computed exactly while an estimate of an unknown stochastic power spectrum remains noisy. The distinction is central to electrophysiological inference.

10.1 Why the raw periodogram is not generally consistent

For a zero-mean Gaussian white-noise series with variance σ^2 and length T, define at a non-DC Fourier frequency

$$I_T(f_k) = \frac{1}{T}\left|\sum_{t=0}^{T-1} X_t e^{-2\pi i k t/T}\right|^2 .$$

At generic positive Fourier frequencies below Nyquist, real and imaginary Fourier coefficients are independent normals with the appropriate variance; hence $2I_T(f_k)/\sigma^2 \sim \chi^2_2$ under this exact white Gaussian model. Consequently the periodogram variance does not vanish as T increases. Averaging approximately independent tapered or segmented estimates reduces variance, at a cost of frequency resolution and possible bias. For K independent identical exponential spectral ordinates, $2K\bar{I}/S \sim \chi^2_{2K}$; Welch overlap and tapered spectra violate exact independence, so their effective degrees of freedom need calibration.

10.2 Short-time Fourier transform, wavelets and admissibility

A window w produces the short-time Fourier transform $V_w x(\tau, f) = \int x(t)\overline{w(t-\tau)}e^{-2\pi i f t}\,dt$. Window duration sets an uncertainty trade-off; the display resolution is not a ground-truth measurement of a neural oscillation's instantaneous frequency. For a continuous wavelet ψ, the usual admissibility condition is

$$C_\psi = \int_{\mathbb{R}} \frac{|\Psi(f)|^2}{|f|}\,df < \infty.$$

It requires zero mean for sufficiently regular wavelets, so an uncorrected finite-frequency Morlet wavelet is only approximately admissible; a DC correction makes the distinction explicit. Reconstruction follows with the appropriate wavelet normalization. Discrete wavelet frames require bounds $A\|x\|^2 \le \sum_j |\langle x, \psi_j\rangle|^2 \le B\|x\|^2$ for positive finite A, B.

10.3 Coherence, phase consistency and finite samples

Even when the population coherence is zero, the squared magnitude of a finite-sample coherence estimate is nonnegative and generally biased upward. A simple idealized average of K independent complex Gaussian spectral replicates gives, under zero population coherence, an estimator with $\mathbb{E}[\widehat{C}] \approx 1/K$; this is *not* a universal threshold for overlapped, tapered or strongly dependent data. For iid spike phases $\phi_1, \ldots, \phi_N$, the usual phase-locking value (PLV) is $|N^{-1} \sum_j e^{i\phi_j}|$. Even uniformly distributed phases produce a positive finite-sample value. The pairwise phase consistency (PPC)

$$\widehat{\mathrm{PPC}} = \frac{1}{N(N-1)} \sum_{j \neq k} \cos(\phi_j - \phi_k)$$

has zero expectation for independent uniform phases, provided the sampling and independence conditions hold. Surrogates made by Fourier phase randomization typically preserve individual Fourier magnitudes; iterative amplitude-adjusted Fourier transforms (IAAFT) approximately preserve both the marginal amplitude distribution and a target power spectrum. Neither automatically preserves cross-channel dependence or experimental trial structure.

Example 10.1 (What is—and is not—implemented)**.** `welch_psd.m`, `multitaper_psd.m`, `wavelet_coherence.m` and `pairwise_phase_consistency.m` produce numerical estimators. A scientific significance statement requires effective degrees of freedom, trial/block design and an appropriate null; it is not guaranteed by the function name.

Chapter 11

Sampling, spectral concentration, and random-matrix theorems

The following classical theorems connect continuous-time measurement, spectral estimation, and high-dimensional statistical models. They are presented here immediately after Fourier and spectral foundations rather than as a revision-history appendix; each requires its precise hypotheses.

11.1 Nyquist–Shannon sampling

Theorem 11.1 (Classical bandlimited reconstruction)**.** *For a finite-energy continuous-time signal with Fourier support inside* $[-B, B]$ *Hz, an ideal uniform, noiseless sampling sequence at* $F_s > 2B$ *determines the signal. With* $\operatorname{sinc}(u) = \sin(\pi u)/(\pi u)$*, reconstruction has the form*

$$x(t) = \sum_{n \in \mathbb{Z}} x(n/F_s) \operatorname{sinc}(F_s t - n),$$

in the suitable convergence sense.

Application: set anti-aliasing and acquisition requirements. Real filters have finite rolloff; extracellular spikes are not literally compactly bandlimited.

11.2 Bedrosian's theorem

Theorem 11.2 (A sufficient spectral-separation case)**.** *If* a *has Fourier support in* $[-A, A]$ *and* c *is supported outside* $(-A, A)$*, with the required transform and product hypotheses, then* $\mathcal{H}(ac) = a\,\mathcal{H}(c)$*.*

For $c(t) = \cos(2\pi f_0 t)$ with $f_0 > A$, an ideal positive envelope $a(t)$ is recovered from the analytic signal of $a(t)c(t)$. The nonoverlap condition is sufficient, not a general necessary and sufficient criterion. PAC remains susceptible to waveform shape, transients, nonstationarity and filtering [1, 9].

11.3 Wiener–Khinchin theorem

Theorem 11.3. *A weakly stationary second-order process has an autocovariance and a corresponding nonnegative spectral measure. When densities exist and inversion is justified,*

$$S_x(f) = \int_{\mathbb{R}} R_x(\tau) e^{-2\pi i f \tau}\, d\tau, \qquad R_x(\tau) = \int_{\mathbb{R}} S_x(f) e^{2\pi i f \tau}\, df.$$

This supports power spectral estimation; it does not guarantee a unique oscillatory/aperiodic decomposition [10, 12].

11.4 Slepian concentration and DPSS tapers

For a length-N taper v and normalized half-bandwidth $0 < W < 1/2$, define the concentration ratio

$$\lambda(v) = \frac{\int_{-W}^{W} |\sum_{n=0}^{N-1} v_n e^{-2\pi i f n}|^2\, df}{\int_{-1/2}^{1/2} |\sum_{n=0}^{N-1} v_n e^{-2\pi i f n}|^2\, df}.$$

Theorem 11.4. *Eigenvectors of the discrete spectral-concentration operator maximize this ratio successively under orthogonality constraints. Its eigenvalues lie in* $[0, 1]$.

Application: `arsenal/path/multitaper_psd.m` uses MATLAB DPSS and averages tapered periodograms, without adaptive taper weighting.

11.5 Johnson–Lindenstrauss lemma

Theorem 11.5 (Finite-set projection)**.** *A suitable normalized random linear map to dimension* $m = O(\varepsilon^{-2} \log P)$ *approximately preserves all pairwise Euclidean distances between* P *fixed points with high probability; constants also depend on the desired failure probability.*

A random dense projection is *not* equivalent to sampling a random subset of neural coordinates. Coordinate sampling needs additional coherence, leverage and sampling assumptions. This theorem does not alone certify reconstruction of neural manifolds.

11.6 Marcenko–Pastur and Tracy–Widom

Theorem 11.6 (Marcenko–Pastur law, standard isotropic model)**.** *For large sample covariance matrices from iid isotropic observations with ratio* $N/T \to \gamma$*, the continuous limiting spectrum is supported on* $[\sigma^2(1 - \sqrt{\gamma})^2, \sigma^2(1 + \sqrt{\gamma})^2]$*; when* $\gamma > 1$ *there is also an atom at zero.*

Theorem 11.7 (Tracy–Widom edge fluctuation)**.** *For suitable real Wishart ensembles, correctly centered and scaled largest-eigenvalue fluctuations at the upper edge follow a Tracy–Widom limit.*

Neither theorem provides an exact universal noise threshold for time-correlated, heteroscedastic, or preprocessed neural data. Edge fluctuations may occur on either side of the limiting edge.

11.7 Jensen's inequality

Theorem 11.8. *If ϕ is convex and the relevant expectations exist, $\phi(\mathbb{E}X) \le \mathbb{E}[\phi(X)]$.*

An elementary dimensionality bound follows instead directly from Cauchy–Schwarz:

$$1 \le D_{\mathrm{PR}} = \frac{(\sum_{i=1}^{r} \lambda_i)^2}{\sum_{i=1}^{r} \lambda_i^2} \le r, \quad \lambda_i > 0.$$

Jensen's more direct role in this textbook is the derivation of variational lower bounds and expectation-maximization, given in the GLM and optimization chapter. Inferring a structural biological dimensionality bound requires additional assumptions.

11.8 Schoenberg positive-definite radial kernels

Theorem 11.9 (Powered exponential, all Euclidean dimensions)**.** *For $\alpha > 0$, the kernel $k(x, y) = \exp(-\alpha\|x - y\|^q)$ is positive definite on all Euclidean dimensions for $0 < q \le 2$; this all-dimension guarantee does not hold for $q > 2$.*

This provides a covariance-valid kernel family for Gaussian processes, but not a proof that a particular model recovers physiological factors. Matern kernels are a different family with their own assumptions.

11.9 Tonelli and Fubini

Theorem 11.10 (Tonelli)**.** *For nonnegative measurable f over a product of sigma-finite measure spaces, iterated integrals can be interchanged, possibly with value $+\infty$.*

Theorem 11.11 (Fubini)**.** *For absolutely integrable measurable f on a product of sigma-finite measure spaces, the integrals can be interchanged and have finite common value.*

These theorems justify integrals used in spike-count expectations and latent-variable models only when their hypotheses apply.

Chapter 12

Counting processes, spike trains, and information

A spike train is a set of event times or, more generally, a random locally finite counting measure. Write $N((a, b])$ for the number of events in an interval. A conditional intensity $\lambda(t \mid \mathcal{H}_t)$ describes the conditional instantaneous event rate given the preceding history $\mathcal{H}_t$, if such a predictable intensity exists.

Result 71: Kolmogorov extension theorem. *Precise setting:* A consistent collection of finite-dimensional probability distributions defines a stochastic process on a suitable product measurable space.

Why it matters here: Build process models from their joint laws across finite time sets.

Result 72: Campbell's theorem. *Precise setting:* For a point process with mean measure Λ and nonnegative measurable h, $\mathbb{E}[\sum_{t_i} h(t_i)] = \int h \, d\Lambda$; signed integrable variants require absolute integrability.

Why it matters here: Compute expected spike-triggered sums and event statistics.

Result 73: Poisson superposition and thinning. *Precise setting:* Independent Poisson processes superpose to a Poisson process with the sum intensity; independent thinning with valid retention probability yields the corresponding Poisson intensity.

Why it matters here: Generate spike-train simulation benchmarks and control models.

Result 74: Doob–Meyer decomposition. *Precise setting:* A suitable integrable counting submartingale admits a predictable compensator $A(t)$, and $N(t) - A(t)$ is a martingale; when absolutely continuous, $A(t) = \int_0^t \lambda(s) \, ds$.

Why it matters here: Connect conditional intensity estimation and martingale residuals.

Result 75: Time-rescaling theorem. *Precise setting:* Under a correctly specified continuous conditional intensity and regularity of event times, transformed inter-event compensator increments are iid Exp(1) (with standard caveats for boundaries and conditioning).

Why it matters here: Construct goodness-of-fit diagnostics for spike-train GLMs.

Result 76: Conditional point-process likelihood. *Precise setting:* For a simple point process with conditional intensity λ on a finite observation interval, under standard domination assumptions its log-likelihood, up to reference terms, is $\sum_i \log \lambda(t_i) - \int \lambda(t) \, dt$.

Why it matters here: Derive maximum-likelihood fitting for spiking models.

Result 77: Palm theory and Slivnyak's theorem. *Precise setting:* For homogeneous Poisson processes, a reduced Palm process at a specified event has the law of the original process; more general Palm laws need not have this property.

Why it matters here: Understand event-centered descriptions and null spike correlations.

Result 78: Gibbs' inequality. *Precise setting:* For probability measures P, Q with $P \ll Q$, $D_{\mathrm{KL}}(P\|Q) \geq 0$, with equality if $P = Q$ almost everywhere.

Why it matters here: Understand KL-based divergence estimators and likelihood objectives.

Result 79: Data-processing inequality. *Precise setting:* For Markov chain $X \to Y \to Z$, $I(X;Z) \leq I(X;Y)$ whenever mutual information is defined.

Why it matters here: Characterize information lost through feature extraction or decoding.

Result 80: Fano's inequality. *Precise setting:* For a discrete class variable with $K \geq 2$ classes, entropy and classification error obey $H(X \mid \widehat{X}) \leq h_2(P_e) + P_e \log(K-1)$, with a consistent logarithm base.

Why it matters here: Relate decoder error to information; do not confuse with the spike-count Fano *factor*.

Example 12.1 (Count likelihood and binning)**.** With constant intensity λ on interval length Δ, $N_\Delta \sim \mathrm{Poisson}(\lambda\Delta)$. Then $\log p(n \mid \lambda) = n\log(\lambda\Delta) - \lambda\Delta - \log(n!)$. A binned GLM approximates continuous-time spiking only relative to its chosen bin width and assumptions about multiple spikes per bin.

Interpretation warning 12.2. A time-rescaling test diagnoses discrepancies from the hypothesized conditional intensity; a passed test is not proof of a unique physiological mechanism. Model misspecification can also arise from latent common drive or refractory dynamics.

Chapter 13

Renewal processes, spike-train spectra, and bin limits

A spike train may be treated as a random counting measure $n(t) = \sum_j \delta(t - t_j)$, where the Dirac deltas are *distributions*, not finite-amplitude voltage spikes. This representation connects point processes to Fourier spectra and to continuous-time LFP shot-noise models.

13.1 How Bernoulli bins approach a Poisson process

Partition a fixed duration T into $m = T/\Delta$ bins, each with independent event probability $p_\Delta = \lambda\Delta$. Then $N_T \sim \text{Binomial}(m, p_\Delta)$, with

$$\mathbb{P}(N_T = 0) = (1 - \lambda T/m)^m \longrightarrow e^{-\lambda T}.$$

The binomial probability generating function likewise converges to $\exp[\lambda T(s-1)]$, proving convergence of the *total count* to a Poisson distribution. Independence of disjoint bins and consistency across intervals give the homogeneous Poisson-process limit. A form of Le Cam's inequality bounds total-variation error in Poisson approximations by a constant multiple of $\sum_i p_i^2$; hence fixed T and $p_i = \lambda\Delta$ make this error vanish as $\Delta \to 0$. This approximation is not a guarantee for refractory or bursty neuronal processes.

13.2 Renewal variability

Suppose interspike intervals $S_1, S_2, \ldots$ are iid, strictly positive, with mean m and finite variance v. Their coefficient of variation is $CV^2 = v/m^2$. Under standard renewal limit conditions, for long observation windows T the spike count satisfies

$$\mathbb{E}[N(T)] \sim T/m, \qquad \text{Var}[N(T)] \sim (v/m^3)T, \qquad \frac{\text{Var}\, N(T)}{\mathbb{E}N(T)} \longrightarrow CV^2.$$

Thus *long-window* count Fano factor and ISI CV^2 are related for a renewal model. Nonrenewal history, trial mixing, latent gain, short windows and nonstationarity can break the empirical approximation.

13.3 Bartlett spectra and shot-noise LFP

For a stationary point process, the Bartlett spectrum is the spectrum of its centered event-counting measure, including the shot-noise contribution. A homogeneous Poisson train of rate λ has flat continuous spectral density $S_{nn}(f) = \lambda$ in a two-sided Hz convention (the DC mean is treated separately). For a stationary renewal process with inter-event characteristic function $\varphi(f) = \mathbb{E}[e^{-2\pi i f S}]$, the nonzero-frequency continuous spectral density takes the form

$$S_{nn}(f) = \lambda \frac{1 - |\varphi(f)|^2}{|1 - \varphi(f)|^2},$$

under the usual nonlattice and regularity assumptions. This shows how refractoriness or regular spiking produces spectral structure without sinusoidal membrane oscillations.

If an idealized LFP component is a linear superposition of spike-triggered kernels, $y(t) = \int h(t - s)\, dN(s)$, its centered spectrum satisfies $S_{yy}(f) = |H(f)|^2 S_{nn}(f)$; independent measurement noise adds its own spectrum. If both n and y derive from the same events, their cross-spectrum is proportional to $H(f)S_{nn}(f)$ with sign/conjugation depending on the cross-spectrum convention. This is an example of *spike–field coherence without a claim of synaptic causation.*

13.4 Hawkes excitation and stability

The classical branching formulation is described by Hawkes [14]. For a linear Hawkes intensity $\lambda(t) = \mu + \sum_{t_j < t} h(t - t_j)$ with $h \geq 0$, $\mu > 0$ and branching ratio $\nu = \int_0^\infty h(s)\, ds < 1$, a stationary version exists under the standard construction, and its mean rate is $\bar{\lambda} = \mu/(1 - \nu)$. When $\nu \geq 1$, this stationary finite-mean construction fails in the usual nonnegative linear model; the statement does not apply without alteration to inhibitory or nonlinear point-process intensities.

Example 13.1 (Arsenal interpretation)**.** Compare `spike_count_fano.m`, `spike_field_coherence.m` and `spike_field_ppc.m` on Poisson, renewal, and latent-rate simulations. A function operating on binned counts or spike phases does not itself implement a continuous-time Bartlett spectral estimator.

Chapter 14

Exponential families, GLMs, and constrained optimization

The Poisson model in `poisson_spike_glm.m` can be derived from a probabilistic model, rather than treated as an unexplained numerical recipe.

14.1 Poisson counts as an exponential family

For count $Y \in \{0, 1, \ldots\}$ and positive mean μ,

$$p(y \mid \mu) = \exp(y \log \mu - \mu - \log(y!)).$$

Its natural parameter is $\eta = \log \mu$. For a row of covariates $x_i^\top$, exposure duration $\Delta_i > 0$, and regression coefficients β, define $\mu_i = \Delta_i \exp(x_i^\top \beta)$. In matrix form with design X,

$$\ell(\beta) = \sum_i [y_i(x_i^\top \beta + \log \Delta_i) - \Delta_i e^{x_i^\top \beta} - \log(y_i!)].$$

Proposition 14.1 (Concavity of the Poisson log likelihood)**.** *For fixed nonnegative counts and strictly positive exposures,*

$$\nabla \ell = X^\top (y - \mu), \qquad \nabla^2 \ell = -X^\top W X, \quad W = \operatorname{diag}(\mu_i) \succ 0.$$

Thus ℓ is concave; if X has full column rank, it is strictly concave at every finite coefficient vector.

Proof. Differentiate the exponential using

$$\frac{\partial \mu_i}{\partial \beta} = \mu_i x_i.$$

For nonzero v, $-v^\top \nabla^2 \ell v = \sum_i \mu_i (x_i^\top v)^2$, positive under full rank. □

Strict concavity is not existence of a finite maximizer. For all-zero counts and an unpenalized intercept, likelihood increases as the intercept tends to $-\infty$.

14.2 IRLS as Fisher scoring

Set $\eta = X\beta + \log \Delta$ and $\mu = e^{\eta}$. Newton scoring solves, at the current iterate,

$$(X^{\top} W X)\, \delta = X^{\top}(y - \mu), \qquad \beta_{\text{new}} = \beta + \delta.$$

This is equivalent to weighted least squares on the working response $z = \eta + (y - \mu)/\mu$, with the exposure offset kept fixed. For the canonical Poisson family, the observed negative Hessian agrees with expected Fisher information. A quadratic ridge penalty adds $-\lambda P \beta$ to the score and λP to the negative Hessian. If the algorithm *clips* η, or lacks line search, the numerical iterations need not monotonically improve the original unclipped likelihood. This is relevant to the present IRLS draft.

14.3 Jensen, the ELBO, and EM monotonicity

For latent variable Z and any density $q(z)$ supported where the joint model is positive, Jensen gives

$$\log p_\theta(y) = \log \int q(z) \frac{p_\theta(y, z)}{q(z)}\, dz \geq \mathcal{L}(q, \theta) := \mathbb{E}_q[\log p_\theta(y, Z)] - \mathbb{E}_q[\log q(Z)].$$

The identity $\log p_\theta(y) = \mathcal{L}(q, \theta) + D_{\mathrm{KL}}(q \| p_\theta(z \mid y))$ shows exactly when the lower bound is tight. Exact EM alternates an E-step setting q to the posterior and an M-step not decreasing the ELBO. This gives nondecreasing observed-data likelihood, but *not* a global optimum. For phase–amplitude coupling, the Tort modulation index is a normalized KL divergence between a binned phase-dependent amplitude distribution and the uniform distribution; positivity comes from Gibbs' inequality, while binning and surrogates are additional statistical decisions.

14.4 KKT and alternating least squares

For differentiable convex f with inequality constraints $g_j(\beta) \leq 0$ and equality constraints $h_k(\beta) = 0$, KKT conditions (with appropriate constraint qualifications) require stationarity, feasibility, nonnegative multipliers and complementary slackness. Sparse Lasso penalties require subgradients rather than ordinary derivatives at zero. For a CP tensor decomposition, exact minimization over a factor block cannot increase the unpenalized objective *of that specified block problem*; alternating such steps does not make the multi-factor objective convex or guarantee global optimality. The exploratory `tensor_cp_als.m` includes ridge terms and does not track a monotonic objective; a monotonicity claim for that code must be tested, not assumed.

Chapter 15

Population geometry, detectability, and identifiability

A low-dimensional fit is not necessarily a detectable or uniquely identified neural mechanism. This chapter separates three questions: *Is a component present in the data? Can the estimator recover its direction? Does that direction have a unique biological meaning?*

15.1 Beyond the Marchenko–Pastur noise edge

Consider an idealized population covariance $\Sigma = I + \theta vv^\top$, with $\|v\| = 1$, $\theta > 0$, iid Gaussian observations, and $N/T \to \gamma \in (0, \infty)$. In the standard spiked-Wishart model, the top population eigenvalue is $1 + \theta$ and the sample-noise edge is $(1 + \sqrt{\gamma})^2$.

Theorem 15.1 (BBP transition, rank-one covariance case)**.** *When $\theta \le \sqrt{\gamma}$, the leading sample eigenvalue approaches the upper noise edge and the sample top direction does not have asymptotically nonzero alignment with v. When $\theta > \sqrt{\gamma}$, the sample leading eigenvalue converges to $(1 + \theta)(1 + \gamma/\theta)$, strictly above the edge, and asymptotic alignment is nonzero.*

The result belongs to the spiked-covariance phase-transition family [2]. It depends on the iid isotropic sampling model. Temporal dependence, heterogeneous neuron variances, preprocessing, and correlated noise modify its calibration. A detectable statistical spike is not automatically an identified cell assembly or an anatomical network.

15.2 Shrinkage is part of covariance estimation

For sample covariance S of N variables and a target T_0 (often proportional to the identity), a linear shrinkage estimator is $\widehat{\Sigma} = (1 - \alpha)S + \alpha T_0$. The optimal α depends on the loss, assumptions, and estimation method. Ledoit–Wolf-type procedures estimate shrinkage to reduce high-dimensional covariance risk [15]; James–Stein concerns shrinkage of normal means in its classical form. Neither is a license to tune hyperparameters on held-out test trials.

15.3 Nonuniqueness of latent coordinates

In Gaussian factor analysis, $y = \Lambda z + \epsilon$ with $z \sim \mathcal{N}(0, I)$, the rotation $\Lambda' = \Lambda Q$, $z' = Q^\top z$ for orthogonal Q leaves the observable covariance $\Lambda\Lambda^\top + \Psi$ unchanged. The latent axes are

therefore not uniquely labeled. For a linear state-space model,

$$x_{t+1} = Ax_t + Bu_t + w_t, \qquad y_t = Cx_t + v_t,$$

any invertible T changes coordinates to $A' = TAT^{-1}$, $B' = TB$, $C' = CT^{-1}$, with transformed process-noise covariance, while preserving the measured trajectories. This is *similarity ambiguity*, not an estimator bug. General nonlinear latent-variable models may have even larger coordinate freedom.

15.4 Alignment and comparison of subspaces

For matrices $X, Y \in \mathbb{R}^{n\times k}$ of aligned, centered samples, the orthogonal Procrustes problem is

$$\min_{Q^\top Q = I} \|XQ - Y\|_F^2.$$

Proposition 15.2 (Procrustes solution)**.** *Let $X^\top Y = U\Sigma V^\top$ be an SVD. Then $Q_* = UV^\top$ is an optimal orthogonal alignment.*

Proof. Expanding the squared norm leaves the variable term $-2\,\mathrm{tr}(Q^\top X^\top Y)$. Writing $Q = UZV^\top$, the trace is $\mathrm{tr}(Z^\top \Sigma) \le \sum_i \sigma_i$, maximized by $Z = I$. □

Canonical correlation analysis (CCA) instead finds directions with maximal *cross-view correlation* after whitening both covariance structures. Representational similarity analysis (RSA) and centered kernel alignment (CKA) compare geometries through different invariances. They require carefully matched observations, preprocessing and independent validation sets.

Example 15.3 (Existing code)**.** `population_pca.m` produces an SVD of centered recordings. `participation_ratio.m` summarizes a covariance eigenspectrum. Neither establishes BBP significance, latent-factor uniqueness, or a well-identified inter-area communication subspace without additional modeling and tests.

Chapter 16

Dependent observations, selective inference, and causality

Time bins from one animal are typically correlated. Repeating a test at many channels, lags and frequency bands also creates selection effects. A valid mathematical result must state which dependence and selection operations it covers.

16.1 Variance of a time-series average

For a zero-mean weakly stationary time series with $\gamma(k) = \mathrm{Cov}(X_t, X_{t+k})$, direct expansion gives

$$\mathrm{Var}(\bar{X}_T) = \frac{1}{T}\left[\gamma(0) + 2\sum_{k=1}^{T-1}\left(1 - \frac{k}{T}\right)\gamma(k)\right].$$

This *finite-sample identity* needs no central limit theorem. When $\sum_k |\gamma(k)| < \infty$, the asymptotic variance is $[\gamma(0) + 2\sum_{k\geq 1}\gamma(k)]/T$. For normalized autocorrelation $\rho(k)$, an approximate effective sample size is $T_{\mathrm{eff}} = T/[1 + 2\sum_{k\geq 1}\rho(k)]$, provided that series is meaningful. Negative correlations can make this heuristic exceed T; it must not be blindly clipped or treated as an exact count of independent animals.

16.2 HAC and resampling under dependence

A heteroskedasticity-and-autocorrelation-consistent (HAC) variance estimator uses weighted autocovariances, for example

$$\widehat{\Omega} = \widehat{\gamma}(0) + 2\sum_{k=1}^{L}\left(1 - \frac{k}{L+1}\right)\widehat{\gamma}(k).$$

This is a Bartlett-window Newey–West form in a scalar, mean-estimation setting. Consistency requires hypotheses on stationarity, weak dependence, moments and bandwidth growth; finite samples and nonstationary trials require scrutiny. Block bootstrap preserves within-block temporal structure but is only justified under suitable dependence assumptions and a suitable block length. Randomly permuting every LFP sample is generally *not* an exchangeable null.

16.3 Multiple comparisons and post-selection

Holm's step-down procedure controls family-wise error under valid marginal p-values without an independence assumption. Benjamini–Yekutieli controls FDR under arbitrary dependence using the harmonic correction $c_m = \sum_{j=1}^{m} 1/j$ in its standard formulation; it may be conservative. A maximum-statistic permutation test controls a family of tests if the randomization acts on the units exchangeable under the joint null. Cluster-permutation procedures use adjacency and a cluster-level statistic; rejection does not locate the exact significant onset, offset, or single electrode. After selecting the largest spectral peak and testing it on the same data, ordinary unadjusted p-values generally fail to reflect that selection. Sample splitting, independent replication or mathematically justified selective inference are possible remedies.

16.4 Observational dependence versus intervention

A directed acyclic graph (DAG) encodes a collection of conditional-independence relations under the Markov property. Paths are blocked or unblocked according to *d-separation*. If Z is an appropriate backdoor adjustment set for a proposed causal effect $X \to Y$, under causal sufficiency assumptions relative to the DAG and positivity,

$$p(y \mid \mathrm{do}(x)) = \int p(y \mid x, z) p(z)\, dz.$$

The formula fails for invalid adjustment sets, unmeasured confounding, or violated structural assumptions. In particular, a lagged correlation does not supply the DAG. Faithfulness is an additional assumption (and can fail by parameter cancellation); the common-cause principle motivates but does not uniquely identify latent confounders.

16.5 Transfer entropy and Gaussian Granger

Transfer entropy from X to Y, conditional on selected histories Y^- and Z^-, is a conditional mutual information: $TE = I(Y_t; X^- \mid Y^-, Z^-)$. For jointly Gaussian variables, with unrestricted correct linear conditional predictors and covariance determinants specified consistently,

$$TE_{X \to Y|Z} = \tfrac{1}{2} \log \frac{\det \Sigma_{\text{reduced}}}{\det \Sigma_{\text{full}}} \quad \text{(nats)}.$$

This Gaussian equivalence is established in [3]. A common multivariate Granger measure is the full log ratio; therefore it equals $2TE$ in this Gaussian setting. This is a theorem about *predictive distributions*, not an identification theorem for causal interventions. Spectral Granger additionally requires consistent VAR transfer functions or spectral factorization (e.g., Wilson methods) and model residual checks. The present `granger_causality.m` computes a temporal log-variance ratio, not spectral Granger; the existing `transfer_entropy.m` is a first-order, discrete plug-in conditional-information estimator.

Interpretation warning 16.1. The two MATLAB primitives are exploratory and do not by themselves establish causal identifiability.

Chapter 17

Dynamical systems, inverse problems, and neural manifolds

A state vector $x(t)$ represents a point in a state space. An ordinary differential equation (ODE) $\dot{x} = f(x, t)$ defines how that point moves, provided a solution exists. A vector of measured rates or LFP values is an observation of the state, not necessarily the complete physical state itself.

Result 81: Picard–Lindelof existence and uniqueness. *Precise setting:* If $f(t, x)$ is locally Lipschitz in x and continuous in t on a suitable neighborhood, a unique local solution exists through a given initial condition.

Why it matters here: Determine when deterministic neural state equations define a well-posed trajectory.

Result 82: Gronwall's inequality. *Precise setting:* A nonnegative function bounded by an integral inequality has an exponential comparison bound under its hypotheses.

Why it matters here: Quantify growth of perturbations to trajectories and models.

Result 83: Lyapunov stability theorems. *Precise setting:* A differentiable positive-definite function with suitable nonpositive or negative orbital derivative implies stability or asymptotic stability under the corresponding assumptions.

Why it matters here: Assess recurrent-circuit attractors and locally stable activity.

Result 84: LaSalle's invariance principle. *Precise setting:* In a compact positively invariant region with $\dot{V} \leq 0$, trajectories approach the largest invariant set where $\dot{V} = 0$, subject to standard hypotheses.

Why it matters here: Study convergence when strict energy decrease is unavailable.

Result 85: Hartman–Grobman theorem. *Precise setting:* Near a hyperbolic equilibrium, a smooth autonomous system is locally topologically conjugate to its linearization.

Why it matters here: Interpret local fixed-point geometry while respecting nonlinear effects.

Result 86: Kalman observability rank test. *Precise setting:* The linear time-invariant pair (A, C) with n states is observable exactly when $\operatorname{rank}[C^\top, (CA)^\top, \ldots, (CA^{n-1})^\top]^\top = n$.

Why it matters here: Assess recoverability of latent states from neural measurements.

Result 87: Kalman controllability rank test. *Precise setting:* The linear time-invariant pair (A, B) is controllable exactly when $\operatorname{rank}[B, AB, \ldots, A^{n-1}B] = n$.

Why it matters here: Characterize reachable states under modeled input interventions.

Result 88: Kalman filter optimality for linear-Gaussian models. *Precise setting:* With known linear dynamics, Gaussian initial state and Gaussian white process/measurement noise, Gaussian conditioning yields recursive conditional means and covariances.

Why it matters here: Estimate trajectories from noisy population observations.

Result 89: Takens' delay-embedding theorem. *Precise setting:* For a generic smooth observation of a compact d-dimensional manifold and generic diffeomorphism, sufficiently many delay coordinates (classically $2d + 1$) embed its state manifold.

Why it matters here: Understand the idealized mathematical conditions for reconstructing dynamics from scalar time series.

Result 90: Whitney embedding theorem. *Precise setting:* Every smooth d-dimensional manifold embeds smoothly into $\mathbb{R}^{2d}$ (with refinements and specific hypotheses).

Why it matters here: Motivate low-dimensional state representations, not neuronal sampling guarantees.

Result 91: Implicit function theorem. *Precise setting:* If a differentiable constraint $F(x, y) = 0$ has an invertible derivative in y at a point, it defines y locally as a differentiable function of x.

Why it matters here: Understand local tangent structure of constraint-defined manifolds.

Result 92: Tikhonov regularization convergence results. *Precise setting:* For a bounded linear inverse problem in Hilbert spaces, with proper regularization conditions and parameter choice relative to noise, regularized estimates converge to an appropriate minimum-norm solution as noise vanishes.

Why it matters here: Describe what stable lead-field inversion can recover and when source nonuniqueness remains.

Definition 17.1. A linear state-space model has

$$x_{t+1} = Ax_t + Bu_t + w_t, \qquad y_t = Cx_t + v_t.$$

The dimensions, covariance models, initial distribution, and observation mechanism must be declared before inferential conclusions.

Interpretation warning 17.2. Whitney and Takens do not guarantee that an arbitrary sparse neuron sample recovers the whole circuit manifold. Observability, measurement design, noise, and sampling density are independent concerns. Similarly, a regularized LFP inverse is not a unique physiological current reconstruction without additional constraints.

Chapter 18

Oscillators, bifurcations, and stochastic dynamics

Oscillatory-looking trajectories can emerge from a dynamical system, an externally driven filter, or chance structure in a finite recording. These are mathematically distinct accounts.

18.1 Hopf normal form and an actual periodic attractor

The complex-valued ordinary differential equation $\dot{z} = (\mu + i\omega)z - |z|^2 z$ with $\omega \neq 0$ can be written $z = re^{i\theta}$, giving

$$\dot{r} = \mu r - r^3, \qquad \dot{\theta} = \omega.$$

For $\mu < 0$ the origin attracts nearby trajectories. For $\mu > 0$ the origin repels, while the circle $r = \sqrt{\mu}$ is an attracting periodic orbit. Its period is $2\pi/|\omega|$. This explicit derivation demonstrates a supercritical Hopf-type normal form; deciding whether experimental LFP oscillations arise from an actual Hopf bifurcation requires more evidence.

18.2 Phase reduction and weak coupling

A stable limit-cycle oscillator with phase coordinate θ responds to weak perturbation $u(t)$ to leading order as

$$\dot{\theta} = \omega + Z(\theta) \cdot u(t),$$

where Z is the infinitesimal phase-response curve (PRC). For two weakly coupled oscillators, averaging can produce a phase-difference equation $\dot{\psi} = \Delta\omega + \varepsilon H(\psi)$. Locking requires a stable fixed point of this equation; oscillatory phase locking need not imply anatomical synapses, and non-sinusoidal harmonics can generate apparent PAC.

18.3 jPCA and skew-symmetric dynamics

If $\dot{x} = Mx$ for a real skew-symmetric matrix $M^\top = -M$, then

$$\frac{d}{dt}\|x\|_2^2 = 2x^\top Mx = 0.$$

A real skew-symmetric matrix has orthogonal real normal form consisting of planar blocks $\left(\begin{smallmatrix} 0 & -\omega_j \\ \omega_j & 0 \end{smallmatrix}\right)$ (and possibly zeros); its nonzero eigenvalues occur as imaginary pairs $\pm i\omega_j$. Thus a fitted rotational component conserves Euclidean energy in the ideal autonomous model. The file `jpca_dynamics.m` fits such a skew generator to finite-difference derivatives [7], not a complete state-space model or a proof of physiological cycles.

18.4 Ornstein–Uhlenbeck noise and the Fokker–Planck equation

For an Itô stochastic differential equation $dX_t = a(X_t)dt + b(X_t)dW_t$, a twice differentiable function f obeys Itô's formula

$$df(X_t) = \left(af' + \tfrac{1}{2}b^2 f''\right) dt + bf'\, dW_t.$$

Its density, under regularity and boundary assumptions, evolves by the Fokker–Planck equation $\partial_t p = -\partial_x(ap) + \frac{1}{2}\partial_{xx}(b^2 p)$. For the Ornstein–Uhlenbeck process $dX_t = -\kappa X_t dt + \sigma dW_t$, with $\kappa > 0$, the stationary variance is $\sigma^2/(2\kappa)$ and the stationary autocovariance is $\sigma^2 e^{-\kappa|\tau|}/(2\kappa)$. An exponential autocorrelation is therefore compatible with a stochastic relaxation process and need not imply an intrinsic neuronal pacemaker or periodic oscillator.

Example 18.1 (Simulation check)**.** Simulate a damped OU fluctuation and a noisy Hopf limit cycle at the same sampling frequency. Compare autocorrelation, PSD and phase stability under different recording lengths. A narrow spectral peak alone does not establish the Hopf model.

Chapter 19

Robust estimates, information bias, and numerical stability

Analysis pipelines need both statistical robustness to unusual data points and numerical robustness to finite precision. Neither follows merely from a mathematical formula being valid.

19.1 Median and median absolute deviation

The sample median has finite-sample replacement breakdown point approaching 1/2: fewer than about half the observations cannot drive it arbitrarily far by contamination. For observations $x_1, \dots, x_T$, define MAD $=$ median$_i\,|x_i - \text{median}_j\, x_j|$. For normal errors a common robust scale estimate is $1.4826\,\text{MAD}$, asymptotically calibrated under the Gaussian model. For artifacted spike waveforms, median/MAD thresholding can be more stable than a mean/standard deviation, but physiological artifacts and nonstationarity can contaminate even robust estimates. An influence function formalizes the estimator's infinitesimal response to a point-mass contamination.

19.2 Empirical distribution bounds and time-rescaling

For iid observations with distribution function F, let F_T be the empirical CDF. A Dvoretzky–Kiefer–Wolfowitz bound states for $\epsilon > 0$

$$\mathbb{P}\left(\sup_x |F_T(x) - F(x)| > \epsilon\right) \le 2e^{-2T\epsilon^2}.$$

It implies uniform consistency (a Glivenko–Cantelli conclusion) in the iid setting. The KS check of rescaled interspike intervals uses an exponential or uniform null only when the conditional intensity is correctly specified and relevant transformed intervals satisfy the needed independence assumptions. Repeated fitting and testing on the same data changes calibration.

19.3 Information-theoretic chain rules and finite-sample bias

For discrete variables, the entropy chain rule is $H(X,Y) = H(X) + H(Y \mid X)$ and $I(X;Y \mid Z) = H(X \mid Z) - H(X \mid Y,Z)$. The Gaussian distribution maximizes differential entropy among random vectors with a specified positive-definite covariance, with entropy $\frac{1}{2}\log[(2\pi e)^d \det \Sigma]$. A plug-in entropy estimator from a finite discrete histogram usually has downward bias. With K effectively occupied bins, a leading Miller–Madow-style correction in *nats* is $(K-1)/(2T)$ under its asymptotic sampling assumptions. It does not cure curse-of-dimensionality or guarantee an unbiased information estimator. Continuous estimators such as KSG have different nearest-neighbor assumptions and biases; the `mutual_information.m` implementation should not be described as using KSG without verifying code.

19.4 Conditioning, least squares, and stable algorithms

If a full-rank matrix X has spectral condition number $\kappa_2(X) = \sigma_{\max}/\sigma_{\min}$, then $\kappa_2(X^\top X) = \kappa_2(X)^2$.

Proof. The singular values of X are the square roots of the eigenvalues of $X^\top X$; hence those eigenvalues are σ_j^2 and the ratio of extremes is squared. □

This explains why explicitly solving normal equations can amplify roundoff relative to QR/SVD solutions. A backward-stable algorithm returns an exact answer for a nearby problem; it does not guarantee small forward error for ill-conditioned inputs. The FFT reduces the computation of a DFT from the naive $O(T^2)$ complex arithmetic to $O(T \log T)$ for appropriate factorizable sizes. Algorithmic efficiency does not change spectral-estimator uncertainty.

Example 19.1 (Research reproducibility)**.** When regressing spike history on population counts, report condition numbers, convergence criteria, effective independent units, model selection, bin duration and handling of omitted variables, not just a coefficient array.

Part V

Connections and worked mathematics

Chapter 20

Worked proofs and derivations

We now move from cataloguing results to deriving selected statements. Every example connects a small mathematical argument with a possible analysis of electrophysiological recordings.

20.1 Proving a set identity

Fix arbitrary x. By definition of union, $x \notin A \cup B$ means that $x \notin A$ and $x \notin B$, which means $x \in A^c \cap B^c$. Thus $(A \cup B)^c = A^c \cap B^c$. Equality of sets was proved by equivalence of membership statements.

20.2 Distinguishing relations from functions

Let U be a set of identified units and T an observation interval. Spike occurrences form a relation $R \subseteq U \times T$. The relation is generally not a function $U \to T$ because one unit may have multiple spike times. It does define $F : U \to \mathcal{P}(T)$ by $F(u) = \{t \in T : (u, t) \in R\}$. The codomain $\mathcal{P}(T)$ is the set of subsets of T.

20.3 Cauchy–Schwarz from an orthogonal projection

For vectors x, y in a real inner-product space, $y \neq 0$,

$$0 \leq \left\| x - \frac{\langle x, y \rangle}{\|y\|^2} y \right\|^2 = \|x\|^2 - \frac{\langle x, y \rangle^2}{\|y\|^2}.$$

Rearrangement yields $|\langle x, y \rangle| \leq \|x\| \|y\|$. The complex case inserts conjugates as dictated by the inner-product convention. On centered square-integrable random variables, this yields $|\operatorname{Cov}(X, Y)|^2 \leq \operatorname{Var}(X) \operatorname{Var}(Y)$. Thus $|\rho_{XY}| \leq 1$ for positive variances, while a constant variable gives undefined rather than zero correlation.

20.4 Least squares as orthogonal projection

For $X \in \mathbb{R}^{T \times p}$ with full column rank, minimize $L(\beta) = \|y - X\beta\|_2^2$. Differentiating gives

$$\nabla L(\beta) = -2X^\top (y - X\beta), \qquad X^\top X \widehat{\beta} = X^\top y.$$

The fitted residual $r = y - X\widehat{\beta}$ is orthogonal to every column of X. One can write $\widehat{\beta} = (X^\top X)^{-1} X^\top y$ under the rank assumption; numerically QR/SVD or stable linear solvers are preferable to explicit inversion.

20.5 PCA as a constrained maximization

Center the rows of $X \in \mathbb{R}^{T \times N}$ with $T > 1$ and form $C = X^\top X/(T-1)$. The projected sample variance along unit vector v is $v^\top C v$. With a Lagrange multiplier λ,

$$\mathcal{L}(v, \lambda) = v^\top C v - \lambda(v^\top v - 1), \qquad Cv = \lambda v.$$

By the spectral theorem, maximizing the Rayleigh quotient selects a leading eigenvector; orthogonality constraints yield subsequent components. This proves PCA optimality relative to *sample variance*, not physiological importance.

20.6 Participation-ratio bounds

For r strictly positive covariance eigenvalues, Cauchy–Schwarz gives $(\sum_i \lambda_i)^2 \le r \sum_i \lambda_i^2$. Because all cross products $\lambda_i \lambda_j$ are nonnegative, $(\sum_i \lambda_i)^2 \ge \sum_i \lambda_i^2$. Therefore

$$1 \le \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} \le r.$$

Equality on the right holds when positive eigenvalues are equal; for $r > 1$, the lower bound is approached if one eigenvalue dominates and the remaining ones vanish relative to it.

20.7 Why a Poisson mixture is overdispersed

Let $G > 0$, $\mathbb{E}G = 1$, and $N \mid G \sim \text{Poisson}(\mu G)$. The conditional mean and variance both equal μG. Total expectation yields $\mathbb{E}N = \mu$; total variance gives

$$\text{Var}(N) = \mathbb{E}[\mu G] + \text{Var}(\mu G) = \mu + \mu^2 \,\text{Var}(G).$$

Thus $\text{Var}(N)/\mathbb{E}N = 1 + \mu \,\text{Var}(G)$. This is a model-based result, not proof that actual response variability comes from a particular neural gain mechanism [13].

20.8 Coherence and spectral positive semidefiniteness

Suppose the spectral matrix is

$$S(f) = \begin{pmatrix} S_{xx}(f) & S_{xy}(f) \\ S_{xy}(f)^* & S_{yy}(f) \end{pmatrix} \succeq 0.$$

The determinant must be nonnegative: $S_{xx}S_{yy} - |S_{xy}|^2 \ge 0$. If both auto-spectra are positive, $0 \le |S_{xy}|^2/(S_{xx}S_{yy}) \le 1$. This proves a normalized spectral dependence bound, not biophysical connectivity.

20.9 Conditional point-process likelihood

If $\lambda(t \mid \mathcal{H}_t)$ is a valid predictable conditional intensity and $t_1, \dots, t_n$ are observed events in $(0, T]$, the point-process log-likelihood relative to a specified reference measure is, up to reference terms,

$$\ell = \sum_{i=1}^{n} \log \lambda(t_i \mid \mathcal{H}_{t_i}) - \int_0^T \lambda(s \mid \mathcal{H}_s)\, ds.$$

The event term rewards model intensity at observed spikes; the compensator penalizes excessive predicted spikes. It follows by combining the small-bin event and no-event probabilities under regularity conditions.

20.10 Predictive influence is not direct physical causation

Let z_t be a latent driver and define observed series $x_t = z_t + \epsilon_t$ and $y_t = z_{t-1} + \eta_t$. Past x can help predict y, yet this construction contains no direct physical edge from x to y. Consequently Granger prediction depends on the chosen variables and may reflect unobserved common inputs [4, 6].

Chapter 21

Theorems and the existing MATLAB arsenal

The paths below are checked against the repository's `arsenal/path/` directory at the start of this edition. Each source file is an exploratory primitive rather than a guarantee of complete replication of the published method. Consult `arsenal/METHODS.tsv` for the current implementation statuses.

Mathematical prerequisite	MATLAB source (under arsenal/path)	Analysis and caution
Sampling, convolution, Hilbert and Bedrosian	`bandpass_lfp.m`, `analytic_phase.m`	Signal phase depends on justified filtering; nonstationarity can create apparent PAC.
Wiener–Khinchin, Parseval	`welch_psd.m`, `coherence_spectrum.m`	Finite-window PSD/coherence estimates, not exact population spectra.
Slepian spectral concentration	`multitaper_psd.m`, `multitaper_coherence.m`	DPSS estimates; estimator design and inference require further checks.
Periodic / aperiodic modeling	`aperiodic_slope.m`, `irasa_decomposition.m`	Exploratory methods; fitting a background does not certify an oscillation.
Circular statistics	`circular_mean.m`, `rayleigh_test.m`	Circular inferential assumptions and dependence of phases matter.
Phase-amplitude and phase locking	`modulation_index.m`, `pac_mean_vector_length.m`, `phase_locking_value.m`	Non-sinusoidal harmonics and surrogates need explicit controls.
Randomization inference	`paired_signflip_test.m`, `jitter_spike_surrogate.m`	Exchangeability or an explicit conditional jitter null is required.

Mathematical prerequisite	MATLAB source (under arsenal/path)	Analysis and caution
Spectral theorem and SVD	`population_pca.m`, `assembly_pca.m`	Variance-optimal directions need not identify biological sources.
Effective spectral dimension	`participation_ratio.m`, `eigenspectrum_powerlaw.m`	Participation ratio does not automatically equal manifold dimension.
Low-rank regression	`reduced_rank_regression.m`, `communication_subspace_projection.m`	Cross-validated prediction needed before mechanistic interpretation.
Tensor decomposition	`tensor_cp_als.m`	CP-ALS primitive is not a complete TCA analysis pipeline.
Conditional expectation, variance	`spike_count_fano.m`, `noise_correlation.m`	Count windows and latent gain change sampling interpretations.
Point-process likelihood	`poisson_spike_glm.m`, `spike_history_design.m`	Binned IRLS approximation; missing latent drive and overdispersion checks.
Poisson event statistics	`spike_bin_counts.m`, `peri_event_histogram.m`	Temporal interval conventions and boundaries affect event counts.
Autoregressive prediction, Wold	`granger_causality.m`	Temporal VAR score; not spectral Granger or direct causation.
Linear dynamics / skew symmetry	`jpca_dynamics.m`	Skew-symmetric fit on prepared trajectories; not full published jPCA.
Linear inverse problems	`field_potential_inverse_tikhonov.m`	Regularized estimate, not uniquely identified neural currents.
Mutual information and DPI	`mutual_information.m`, `transfer_entropy.m`	Finite-sample bias, history embedding and discretization matter.

21.1 Theorems without a dedicated implementation

The catalogue also includes Kolmogorov extension, time-rescaling goodness-of-fit, Davis–Kahan perturbation, Takens embedding, and Kalman observability. Gaussian KL divergence and Gaussian conditioning are now derived explicitly in the Gaussian toolkit These are *mathematical prerequisites or future checks*, not claims that their complete scientific imple-

mentations exist in the repository. The presence of `gaussian_process_smoothing.m` must not be interpreted as a full GPFA model.

21.2 Operational contract

Read input shapes and units from the MATLAB function help. Document sample rate, time stamps, channel orientation, boundary conventions, electrode reference, trial alignment, and the meaning of each null hypothesis. Benchmark a method against simulated data of known ground truth before trying to infer physiology from an empirical recording.

21.3 From biological generators to filters and estimates

The physical forward chain has distinct operations: a cable/membrane source model supplies transmembrane currents; a conductivity model with boundary conditions constructs a lead field; `field_potential_forward.m` applies a *supplied* lead field; `bandpass_lfp.m` then uses Butterworth sections and forward–backward noncausal filtering; spectral estimators operate on that processed trace. Inverting a supplied lead field is regularized but not necessarily unique. No software primitive converts a fitted spectral peak directly into a uniquely identified membrane current.

21.4 Model-fitting limitations exposed by the new chapters

`poisson_spike_glm.m` uses a binned canonical Poisson model with clipped linear predictors: deriving the ideal log-concave Poisson objective does not guarantee that every numerical iterate of the implementation maximizes the same unclipped likelihood. `tensor_cp_als.m` is a ridge-regularized CP-ALS primitive; alternating steps are not a global-convergence theorem. `granger_causality.m` returns a temporal predictive log-variance ratio, while `transfer_entropy.m` uses discretized first-order conditional mutual information. The Gaussian equivalence theorem does not claim these distinct implementations have matching finite-sample outputs or identify causal interventions.

Chapter 22

Exercises, counterexamples, and research projects

The following problems are original exercises for the CON^{2}PHYS curriculum. The objective is to prove each mathematical claim first and only then write MATLAB experiments.

Exercise 22.1 (Relations and functions)**.** Define $R \subseteq U \times T$ for two units firing at the same timestamp. Prove that R need not be a function $U \to T$. Construct the corresponding function into $\mathcal{P}(T)$.

Exercise 22.2 (Sampling failure)**.** Sample a 90-Hz sine wave at 100 Hz and identify a lower-frequency sinusoid with the same samples. Explain why a digital filter applied after sampling cannot recover the true generating frequency.

Exercise 22.3 (Hilbert envelope)**.** Construct $x(t) = (1 + 0.3\cos 2\pi 2t)\cos 2\pi 40t$. Calculate the analytic envelope and compare against a signal whose envelope frequencies overlap its carrier spectrum. Explain when the Bedrosian assumption fails.

Exercise 22.4 (Spectral concentration)**.** Generate equal-length noisy signals with an off-bin sinusoid. Compare ordinary periodograms, Welch, and DPSS multitaper estimates using repeated simulations. Vary time-bandwidth product and report bias versus variance.

Exercise 22.5 (Principal subspace perturbations)**.** Generate a diagonal covariance with a controllable eigengap. Perturb it by symmetric noise, compute leading subspace angles, and compare their variation with the Davis–Kahan scaling.

Exercise 22.6 (Random-matrix nulls)**.** Simulate isotropic Gaussian matrices with independent samples, then introduce temporal autocorrelation and heterogeneous channel noise. Compare each empirical eigenvalue histogram with the corresponding idealized Marcenko–Pastur curve.

Exercise 22.7 (Latent gain)**.** Generate counts with $N \mid G \sim \mathrm{Poisson}(\mu G)$, where G is a positive random variable of mean one. Verify the predicted quadratic mean–variance relation and test how a purely Poisson GLM misestimates uncertainty.

Exercise 22.8 (Conditional point-process validation)**.** Fit a simulated point-process intensity with and without spike-history effects. Examine rescaled inter-event intervals. Explain why correctly transformed residuals do not identify a unique mechanism.

Exercise 22.9 (Shared-input Granger counterexample)**.** Simulate an unobserved driver of two time series with different lags. Show that adding measurements of the common source can

change inferred predictive direction or strength. Avoid interpreting this as proof of synaptic connectivity.

Exercise 22.10 (Inverse problem)**.** Construct a lead-field matrix with a nontrivial nullspace. Find two distinct current patterns with identical electrode predictions, then compare Tikhonov solutions for different regularization strengths.

22.1 Suggested cumulative projects

Project I: from set membership to a reliable spike-count representation. Project II: from convolution to LFP spectral density and multitaper estimators. Project III: from covariance to PCA stability and manifold caveats. Project IV: from conditional probability to spike-train GLM validation. Project V: from state-space observability to limits of causal and mechanistic inference.

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